Exercise Set 2.2

Q3. $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \\ 5 & -3 & 6 \end{bmatrix}$ so that $A^T = \begin{bmatrix} 2 & 1 & 5 \\ -1 & 2 & -3 \\ 3 & 4 & 6 \end{bmatrix}$

Now $\det(A) = ?$ and $\det(A^T) = ?$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \\ 5 & -3 & 6 \end{bmatrix} \begin{matrix} 2 & -1 \\ 1 & 2 \\ 5 & -3 \end{matrix}$$

So $\det(A) = [24 - 20 - 9] - [30 - 24 - 6]$

$= -5$ — (1)

Now $A^T = \begin{bmatrix} 2 & 1 & 5 \\ -1 & 2 & -3 \\ 3 & 4 & 6 \end{bmatrix} \begin{matrix} 2 & 1 \\ -1 & 2 \\ 3 & 4 \end{matrix}$

$\det(A^T) = [24 - 9 - 20] - [30 - 24 - 6]$

$= -5$ — (2)

From (1) and (2), it follows that

$\det(A) = \det(A^T)$

Q14. $\begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{vmatrix}$ $R_2 - 5R_1$, $R_3 + R_1$, $R_4 - 2R_1$

$= \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 108 & 23 \end{vmatrix}$ $R_4 - 12R_2 = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 0 & -13 \end{vmatrix}$ $R_4 + 36R_3$

$= 39$

Q17. Given $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$

Now $\begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix} = -(3)(4)\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ $\begin{matrix} \frac{1}{3}R \\ -R_2 \\ \frac{1}{4}R_3 \end{matrix}$

$= -12(-6) = 72$

Q21. $\begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix}$

$= -3\begin{vmatrix} a & b & c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix}$ $-\frac{1}{3}\cdot R_1$

$= -3\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ $R_3 + 4R_2$

$= -3(-6) = +18$

Q23

LHS $= \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ Taking Transpose

$= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix}$ $\begin{matrix} R_2 - R_1 \\ R_3 - R_1 \end{matrix}$

$$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 1 & c+a \end{vmatrix} \begin{matrix} \frac{1}{b-a} R_2 \\ \frac{1}{c-a} R_3 \end{matrix}$$

Note that if $a=b$ or $a=c$

LHS = RHS = 0 so we assumed here that $a \neq b$ and $a \neq c$

$$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & c-b \end{vmatrix} R_3 - R_2$$

$$= (b-a)(c-a)(c-b) = \text{RHS}$$

QED

Q24. b)

$$LHS = \begin{vmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$= -\begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{14} \end{vmatrix} \quad R_{14}$$

$$= +\begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & 0 & 0 & a_{14} \end{vmatrix} \quad R_{23}$$

$$= a_{41} a_{32} a_{23} a_{14} = a_{14} a_{23} a_{32} a_{41}$$

$$= RHS$$

Q26. $\begin{vmatrix} a_1+b_1t & a_2+b_2t & a_3+b_3t \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

$= \begin{vmatrix} a_1-a_1t^2 & a_2-a_2t^2 & a_3-a_3t^2 \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ $R_1 - tR_2$

$= (1-t^2)\begin{vmatrix} a_1 & a_2 & a_3 \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ $\frac{1}{1-t^2} R_1$, assume $t \neq 1$

$= (1-t^2)\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ $R_2 - tR_1$

Q27. $\begin{vmatrix} a_1+b_1 & a_1-b_1 & c_1 \\ a_2+b_2 & a_2-b_2 & c_2 \\ a_3+b_3 & a_3-b_3 & c_3 \end{vmatrix}$

$= \begin{vmatrix} 2a_1 & a_1-b_1 & c_1 \\ 2a_2 & a_2-b_2 & c_2 \\ 2a_3 & a_3-b_3 & c_3 \end{vmatrix}$ $C_1 + C_2$

$= 2\begin{vmatrix} a_1 & a_1-b_1 & c_1 \\ a_2 & a_2-b_2 & c_2 \\ a_3 & a_3-b_3 & c_3 \end{vmatrix}$ $\frac{1}{2} C_1$

$= 2\begin{vmatrix} a_1 & -b_1 & c_1 \\ a_2 & -b_2 & c_2 \\ a_3 & -b_3 & c_3 \end{vmatrix}$ $R_2 - R_1 = -2\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ $= C_2$

Q29. $\begin{vmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{vmatrix}$

$= \begin{vmatrix} -2 & 8 & 1 & 4 \\ 1 & 10 & 6 & 5 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{vmatrix}$ $R_2 + R_1$

$= 0$ Because 2nd and 3rd rows are proportional

Q30. $\begin{vmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{vmatrix}$

$= \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{vmatrix}$ $R_1 + (R_2 + R_3 + R_4 + R_5)$

$= 0$ expansion along 1st row

Q31.

$$M = \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 3 & 6 & -9 \\ 2 & 5 & 0 & 4 & 7 & 5 \\ -1 & 3 & 2 & 6 & 9 & -2 \\ \hline 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & -3 & 8 & -4 \end{array}\right]$$

As $M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ is partitioned so that A and B are square matrices and M is in block triangular form, so that

$\det(M) = \det(A)\det(B)$

Now $\det(A) = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \\ -1 & 3 & 2 \end{vmatrix}$

expand it along 3rd column

$= 0C_{13} + 0C_{23} + 2C_{33}$

$= 2(-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = 2(5-4) = 2$

& $\det(B) = \begin{vmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 8 & -4 \end{vmatrix} = -12$ (triangular)

Thus $\det(M) = 2(-12) = -24$

Q33. To state such a Theorem, first observe that to move the last row of B to first row requires $(n-1)$ interchange operations. Now last row of B will be second row of A. To move this row back to its original position further require $(n-2)$ switchings and so on.

Thus to convert B back to A requires $\frac{n(n-1)}{2}$ interchange operations.

$\therefore \det(B) = (-1)^{\frac{n(n-1)}{2}} \det(A)$.

Q33.

To move B back to A requires $\left[\frac{n}{2}\right]$ operations of interchanging rows. For example, if n is even, then interchanging 1st and last rows, then 2nd and 2nd last rows and so on. And a similar argument can be applied to if n is odd except the middle row will stay at its original position.

Thus $\det(B) = (-1)^{\left[\frac{n}{2}\right]} \det(A)$

Q34. $\begin{vmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{vmatrix}$

$= \begin{vmatrix} a+3b & b & b & b \\ a+3b & a & b & b \\ a+3b & b & a & b \\ a+3b & b & b & a \end{vmatrix}$ $C_1 + (C_2 + C_3 + C_4)$

$= (a+3b) \begin{vmatrix} 1 & b & b & b \\ 1 & a & b & b \\ 1 & b & a & b \\ 1 & b & b & a \end{vmatrix}$ $\frac{1}{a+3b} R_1$

assume $(a \neq -3b)$

$= (a+3b) \begin{vmatrix} 1 & b & b & b \\ 0 & a-b & 0 & 0 \\ 0 & 0 & a-b & 0 \\ 0 & 0 & 0 & a-b \end{vmatrix}$ $R_2 - R_1$, $R_3 - R_1$, $R_4 - R_1$

$= (a+3b)(a-b)^3$